DATA SCIENCE

DATE : 17.11.2025

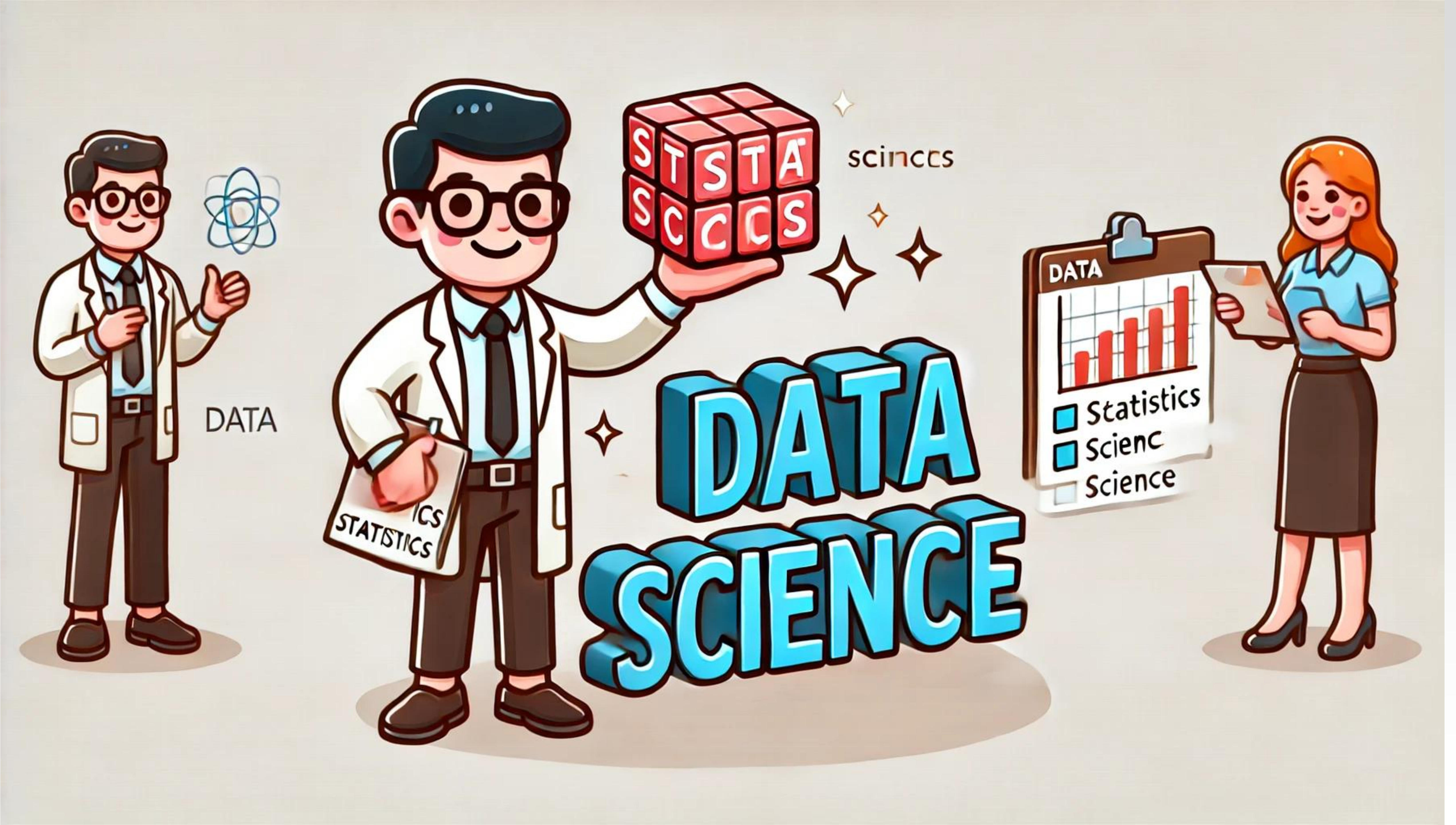
DT/NT : DT

LESSON : STATISTICS I

SUBJECT : INTRO, STATS FOR DS, STATS & DATA TYPES

BATCH : 322

TECHPRO EDUCATION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SCCCS
scincts
DATA
DATA
SCIENCE
DATA
Statistics
Scienc
Science
STATISTICS
ICS

What will we learn in this course?

Part-1

- The meaning of Statistics
  - ➢ Eisenhower Matrix
  - ➢ Characterization
    - ▪ Collection
    - ▪ Analyzing
    - ▪ Visualization
    - ▪ Inference
    - ▪ Presentation
  - ➢ Why Statistics
  - ➢ Importance of Statistics
    - ▪ Data Science vs Statistics
  - ➢ How much Statistics for us
  - ➢ Statistic Types
    - ▪ Descriptive
    - ▪ Inferential
  - ➢ Data Types
  - ➢ Parameters and Statistics
  - ➢ Probability vs Statistics
  - ➢ Level of Measurement
    - ▪ Nominal
    - ▪ Ordinal
    - ▪ Interval
    - ▪ Ordinal

Part-2

- Data Visualization - Graphical Represent
- Patterns
  Center
  - ➢ Spread
  - ➢ Shape
    - ▪ Symmetric
    - ▪ Number of peaks
    - ▪ Skewness
    - ▪ Uniform
  - ➢ Unusual Features
    - ▪ Gaps
    - ▪ Outliers
  - ➢ Frequency Table
    - ▪ Relative Frequency
    - ▪ Cumulative Frequency
  - ➢ Bar Chart
  - ➢ Pie Chart
  - ➢ Histogram
- Populations & Samples
  - ➢ Parameters & Statistics
- Central Tendency (Measure of Centre)
  - ➢ Mean
  - ➢ Median
  - ➢ Mode
- Dispersion (Measure of Spread)
  - ➢ Range
  - ➢ IQR
  - ➢ Standard Deviation
    - ▪ Empirical Rule
  - ➢ Variation

Part-3

- Scatter Plot
  - ➢ Linearity
  - ➢ Slope
  - ➢ Strength
  - ➢ Unusual Features
    - ▪ Clusters
    - ▪ Gaps
    - ▪ Outliers
- Box Plot
  - ➢ Min & Max Values
  - ➢ 1.5*IQR (John Tukey)
- Covariance
- Correlation
  - ➢ Pearson Correlation Coefficient
  - ➢ Correlation – Linear Relationship

➕ Peardeck
➕ Videos
➕ Kahoot
➕ Interview questions
➕ Notebook hands-on practices

What will we learn in this course?

Top 60 Statistics Interview Questions 2024

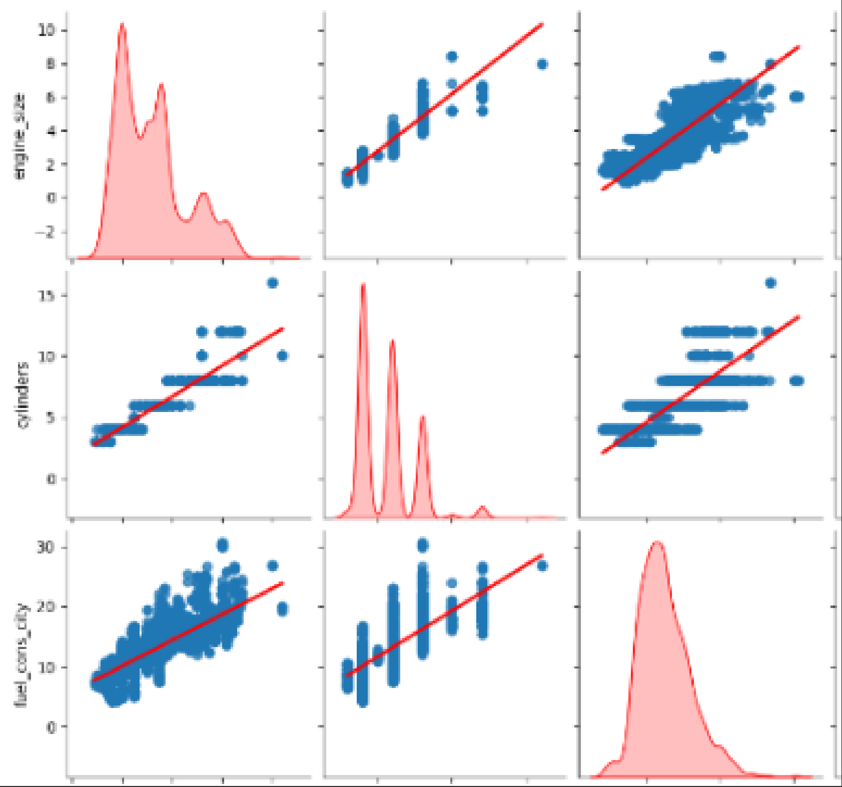
Kahoot!

What will we learn in this course?

3.3 Pairplot

```
sns.pairplot(df, hue = "Survived", palette="Set1");
```

```
# Benim (Ismet) duzenledigim, dagilimlar arasina kirmiz
sns.pairplot(df, kind = "reg", diag_kind = "kde", diag_k
```



3.4 Boxplot

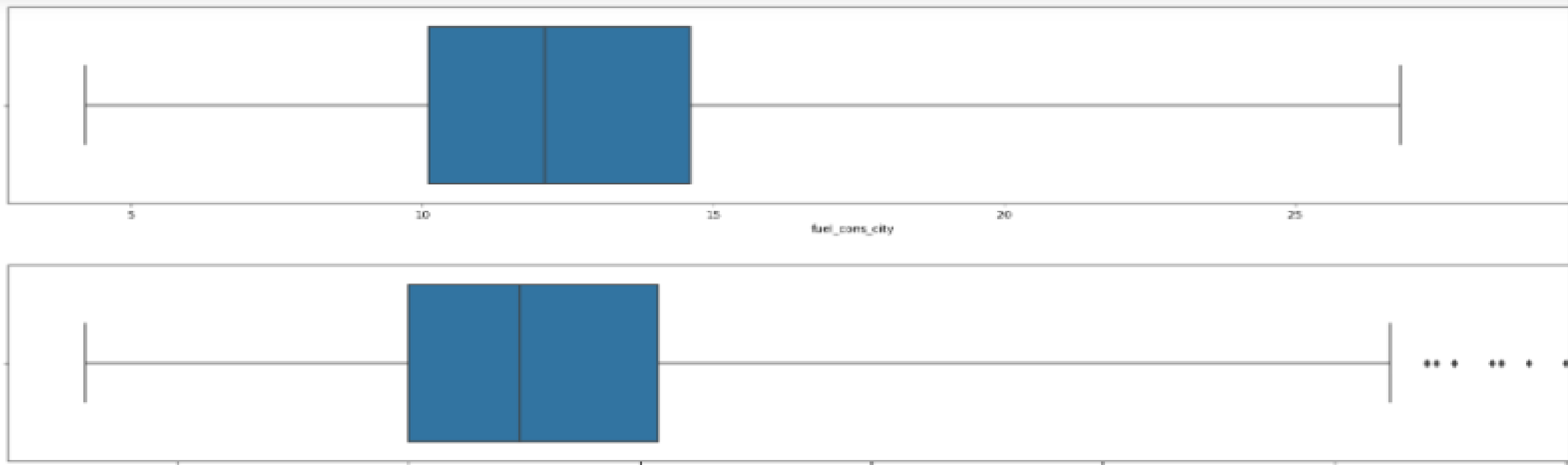
```
sns.boxplot(data=df,y="age",x='income');
```

```
index = 0
plt.figure(figsize=(20,20))
for feature in df.columns:
    if feature != "Survived":
        index += 1
        plt.subplot(3,3,index)
        sns.boxplot(x='Survived',y=feature,data=df)
plt.show()
```

```
# Let's draw boxplots and histplots for checking distributions of features;
index=0
for feature in df.select_dtypes('number').columns:
    index+=1
    plt.figure(figsize=(40,40))
    plt.subplot((len(df.columns)),2,index)
    sns.boxplot(x=feature,data=df, whis=3)

    plt.tight_layout()

    plt.show()
```



Section-1 Content

Content

1. The meaning of Statistics
2. Why Statistics
3. Importance of Statistics
4. How much Statistics for us
5. Statistic Types
6. Data Types
7. Parameters and Statistics
8. Probability vs Statistics
9. Level of Measurements

Have you practiced with the **pre-class** materials that will prepare you for today's lesson?

Fundamentals Statistics

Online Resources

- YouTube [StatQuest] (Funny)
- YouTube [CrashCourse]
- https://www.khanacademy.org/
- Statistics - A Full University Course on Data Science Basics

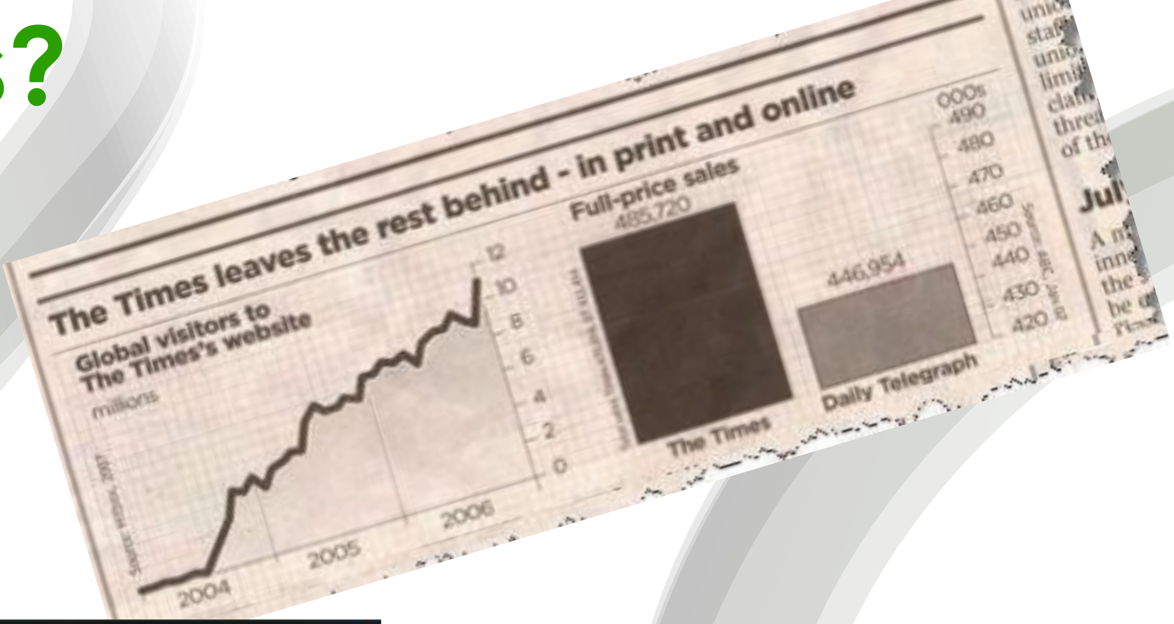
Documents

- LMS page
- PDF Document on Internet

Books:

- Applied Statistics and Probability – Montgomery
- An Introduction to Statistical Learning – James –Witten
- Probability&Statistics - Myers

Why statistics?



Why statistics?

Why statistics?

- ✓ If you say "I know statistics" in an interview, it will increase your offer chance.

- ✓ If you use statistical evidence, figures, and numbers in your presentations, you will get the attention of the audience easily.
- ✓ If you interpret your analysis results by using statistical criteria, your teammates respect you.

Statistics in daily life examples?

Statistics in Daily life examples

Statistics Role In Real Life

- Government
- Emergency Preparedness
- Political Campaigns
- Sports
- Research
- Education
- Prediction
- Predicting Disease
- Insurance
- Financial Market
- Business Statistics

https://medium.com/@john_marsh7/statistics-role-in-real-life-a6ba727e0ad8

Statistics In Our Day-to-Day Life

- Medical Study. Statistics are used behind all the medical study. ...
- Weather Forecasts.
- Quality Testing. A company makes thousands of products every day and make sure that they sold the best quality items.
- Stock Market.
- Consumer Goods.

https://statanalytica.com/blog/uses-of-statistics/

1. What is the statistics?

- Statistics is the study of how to collect, organize, analyze, and interpret numerical information and data.

= Data Engineering

= Data Scientist

- Statistics is both the science of uncertainty and the technology of extracting information from data.

= Data Analytics, BI

- Statistics is used to help us make decisions. This is especially important in health care and public health.

Statistician

Royal Papworth Hospital NHS Foundation Trust

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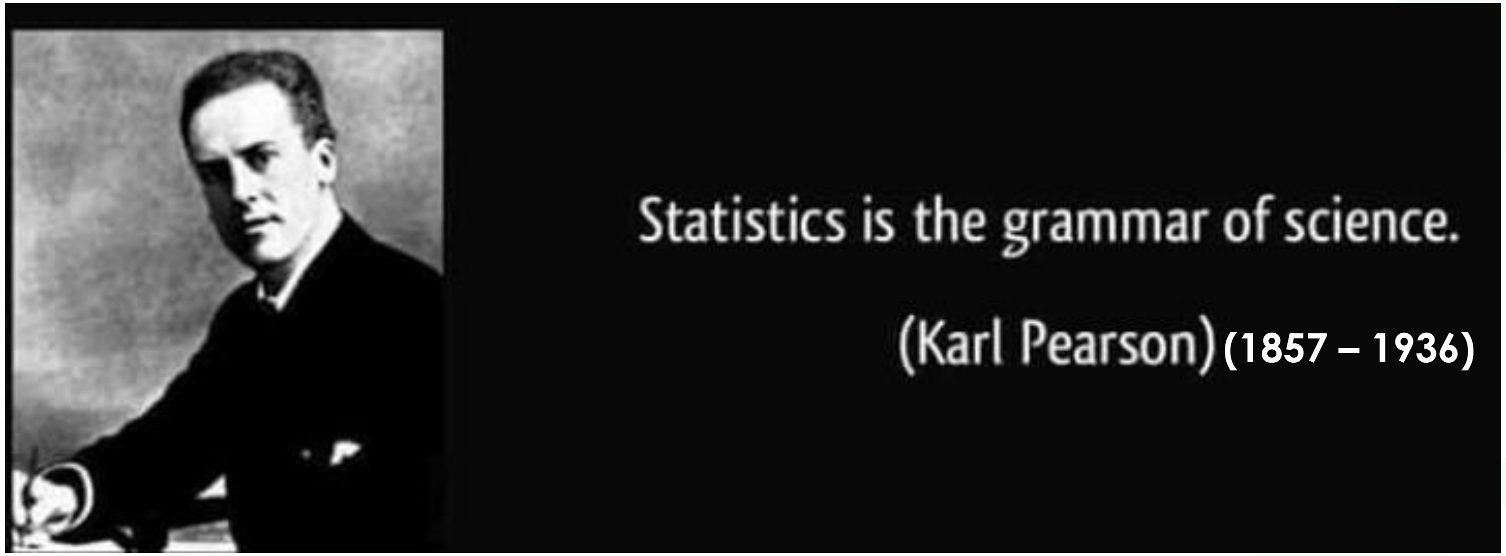
1. What is the statistics?



Statistics as a method:

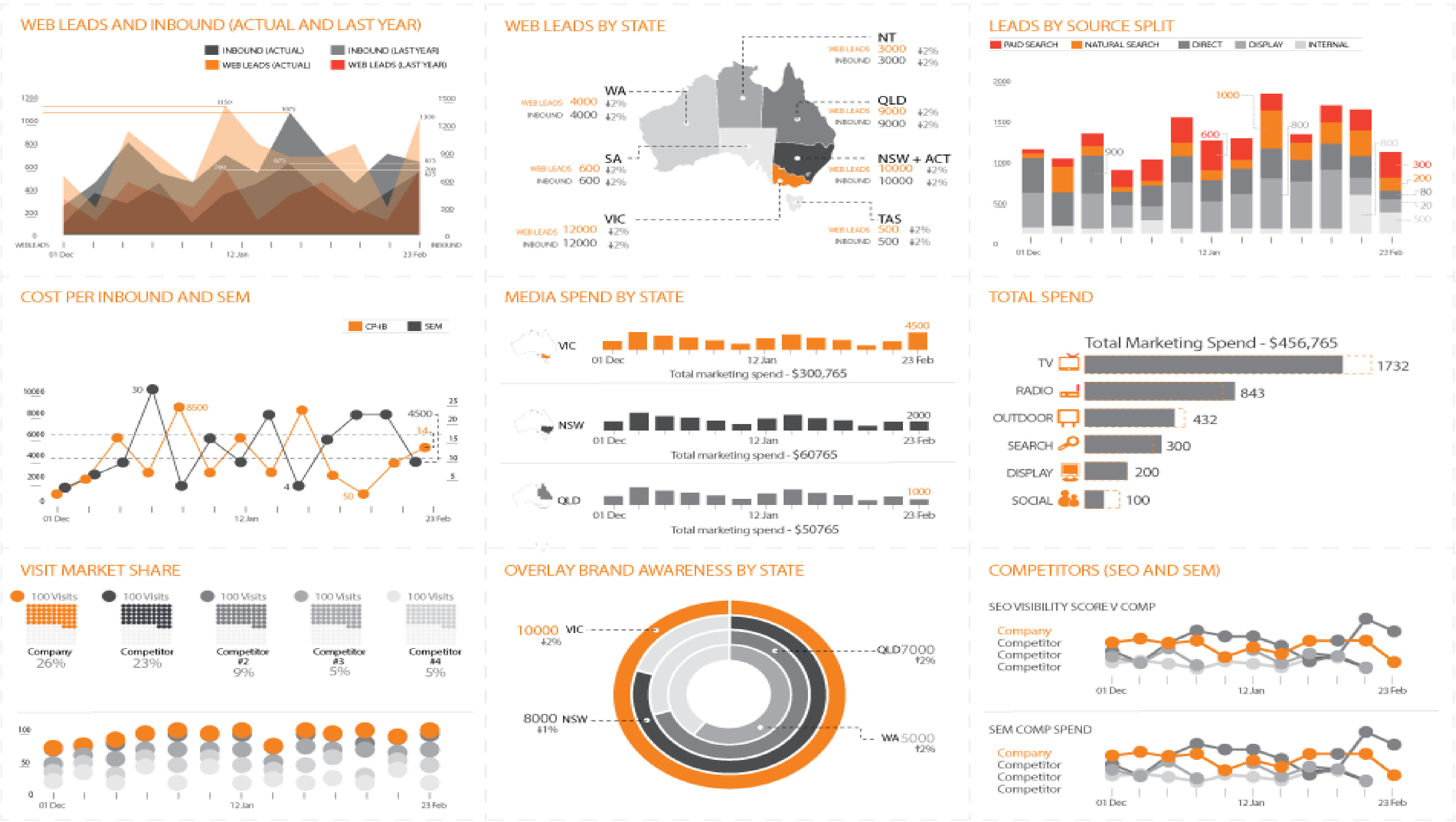
• It is a set of techniques used in **collecting**, **processing**, **analysing** and **interpreting quantitative data** on events that can be subject to statistics.

1. What is the statistics?

What is
statistics?

1. What is the statistics?

Example of Statistics as Dashboard



Next Generation Dashboard

1. What is Statistics

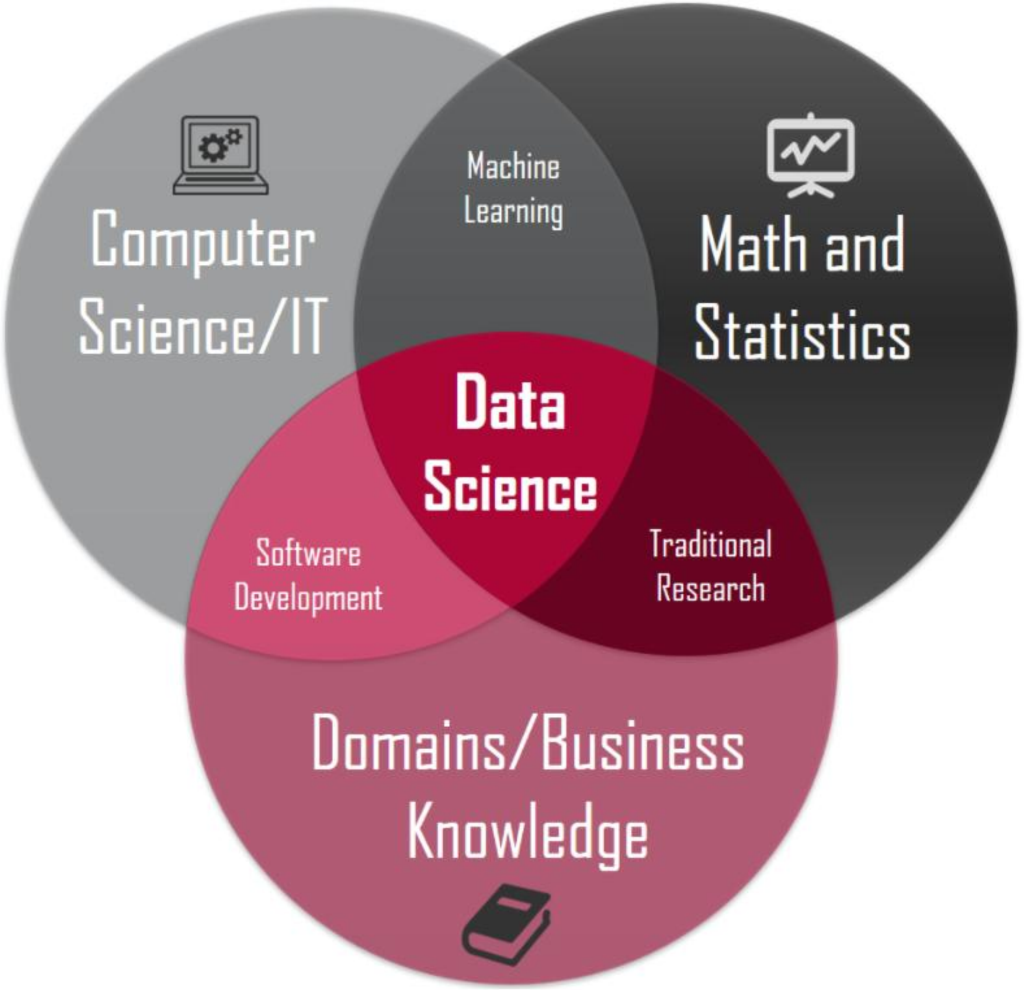
Statistics is all about data

- Collection
- Characterization
- Analysing
- Visualisation
- Inference
- Presentation

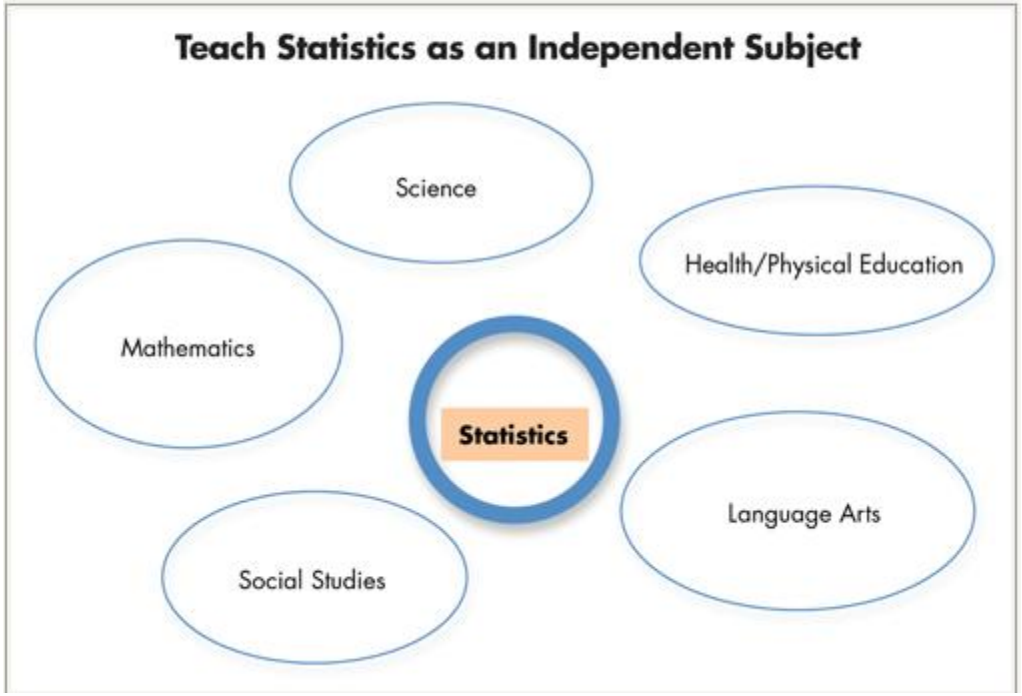
Where is the Statistics in Data Science?

2. Why Statistics

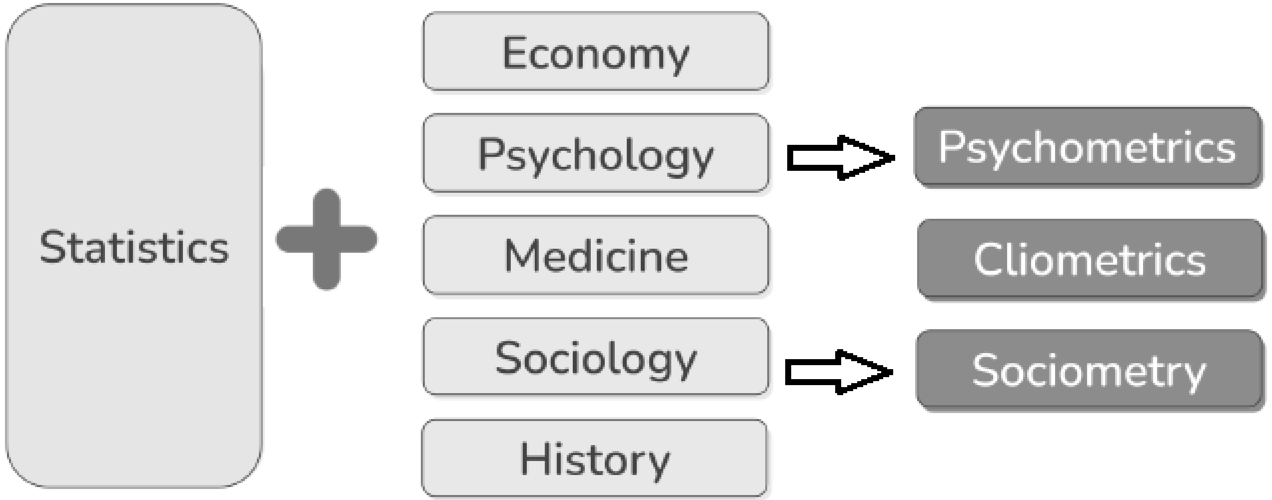
- It helps researchers to explain their findings.
- It helps researchers go beyond the data to more general conclusions.
- Definition of Data Scientist; "**a person who knows more statistics than a programmer and more programming than a statistician**"
- How much Statistics will we learn?



3. Importance of Statistics in Science



https://www.emathzone.com/tutorials/basic-statistics/importance-of-statistics-in-different-fields.html



3. Why statistical knowledge are important in Data Science

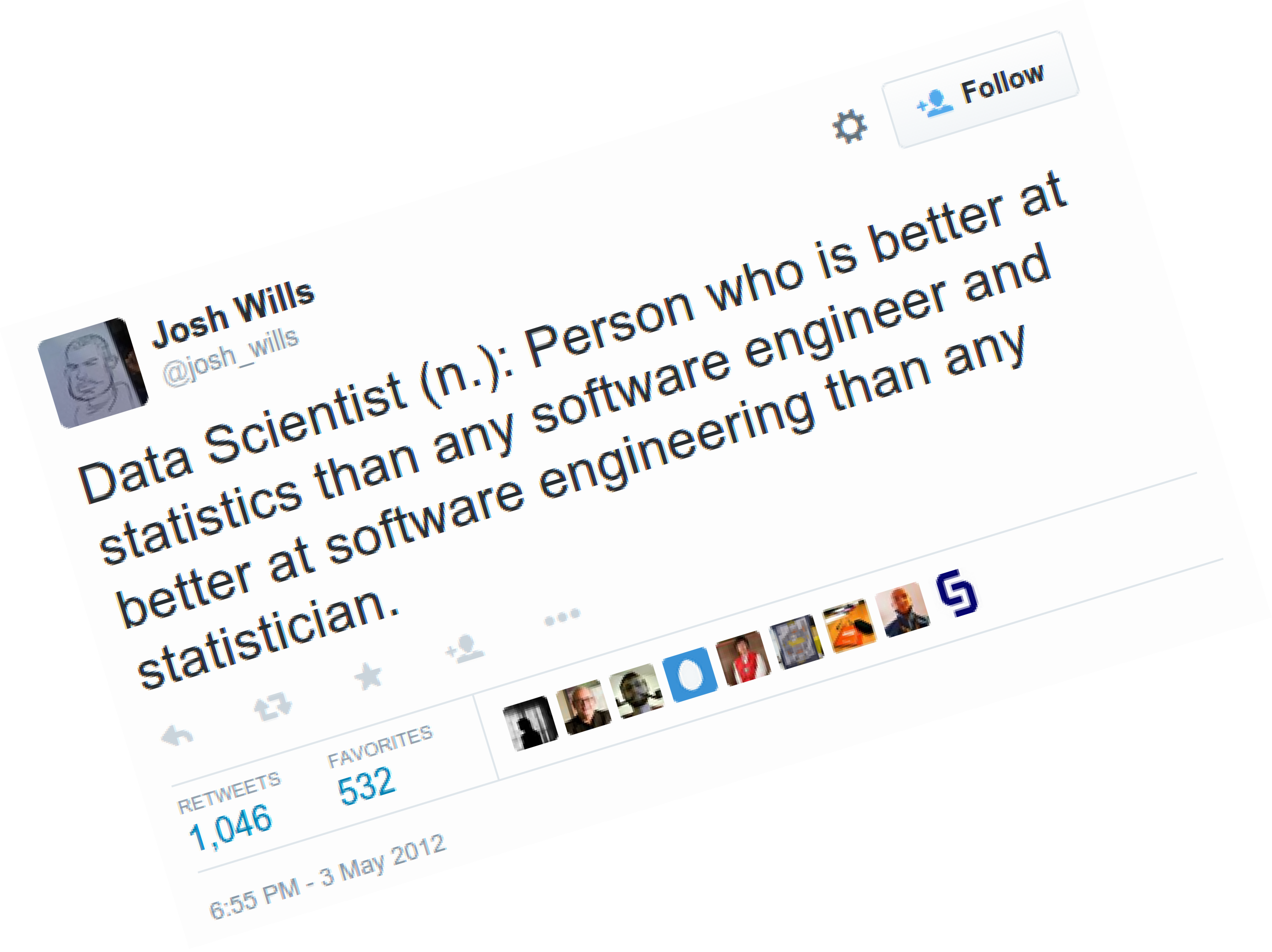
Should I run for the bus?

Which stock should I buy?

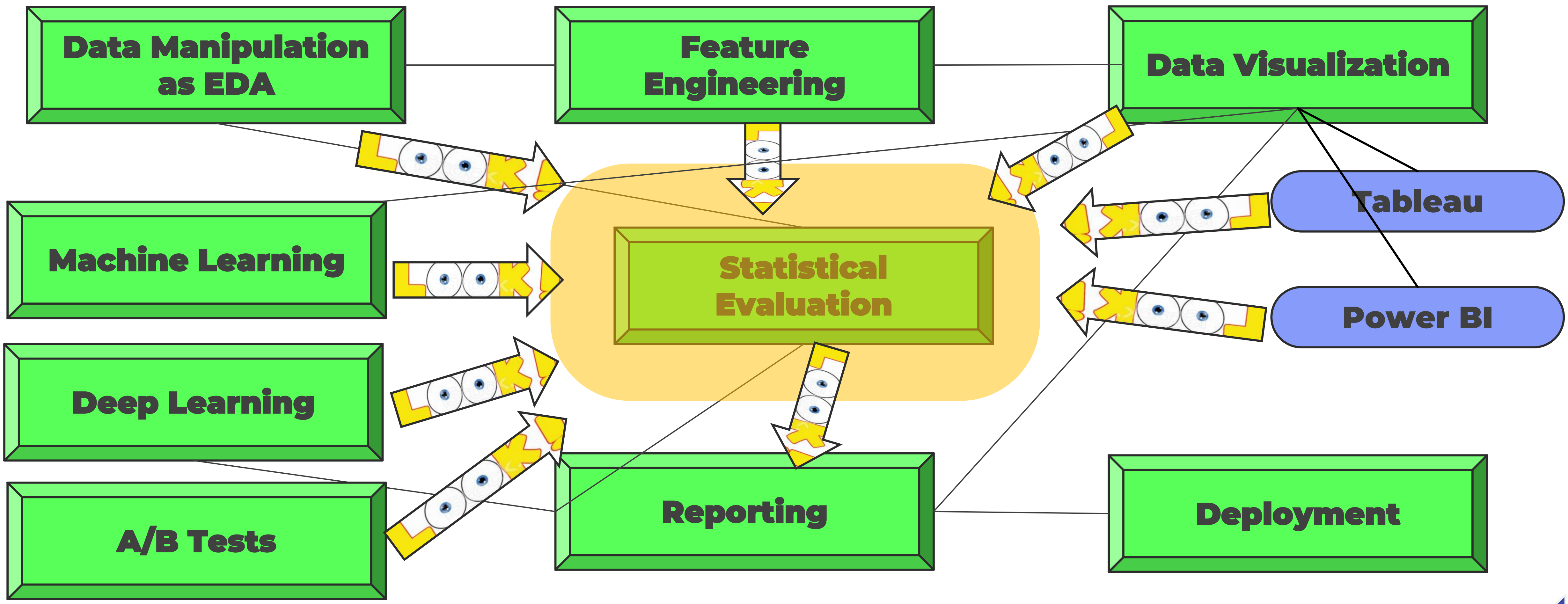
Which products should I list in Website?

Should I take this medication?

Should I have my children vaccinated?



4. Where is the Statistics in Data Science Roadmap



Data Flowing

EDUCATION

Statistical Order

Step One

Collecting data

Step Two

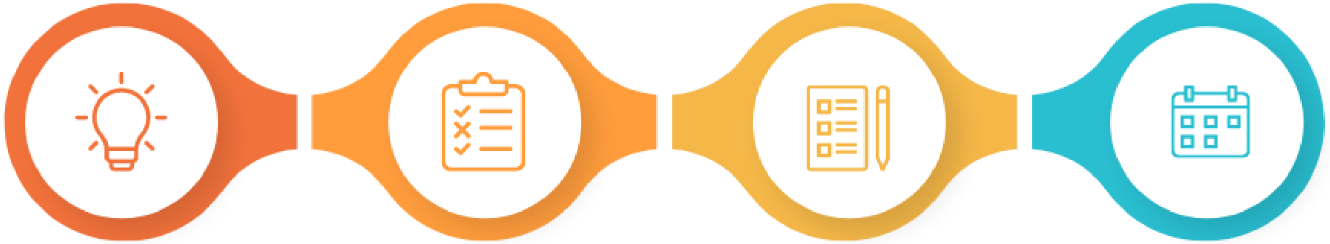
Processing and organizing the collected data

Step Three

Presenting organized data with tables and graphs

Step Four

Statistical analysis and decision



Data Science vs Statistics

Differences

Similarities

- Understanding of mathematics
- investigating problems
- exploratory data analysis
- analyzing **trends**, **paterns**
- creating **forecasts**
- visualizations
- reporting findings to **non-technical users**

Statistics

- one-off reports
- use of SAS programming
- focus on diagnostic plots
- focus on significance testing
- focus on t-tests, ANOVA, and MANOVA, etc.
- more manual data collection (sometimes from surveys)
- usually, you will find Statisticians in the healthcare and economics field
- or more academic settings

Data Science

- automation
- use of SQL querying for data collection
- machine-learning libraries like sklearn and TensorFlow
- use of Python and R programming languages
- deployment of automated models (into an app)
- focus on software engineering practices

DS vs Statistics

DS

A data scientist makes hundreds of decisions every day.

Many of these decisions require a strong foundation in math and statistics.

Data science requires descriptive statistics and probability theory, at a minimum.

DS

- Data science is a multidisciplinary field which uses scientific methods, processes, and systems to extract knowledge from data.
- Data science problems often relate to making predictions and optimizing search of large databases.
- Data scientists tend to come from engineering backgrounds.

Statistician

- Statistics is a mathematically-based field which seeks to collect and interpret quantitative data.
- In contrast, the problems studied by statistics are more often focused on drawing conclusions about the world at large.
- Statisticians are usually trained by math departments.

5. Statistics Types

Statistics

Descriptive Statistics

Summarize the data set.

Measures of Central Tendency:

1. **Mean:** The average of all values.
2. **Median**: The middle value when data is ordered.
3. **Mode**: The most frequently occurring value.

Measures of Variability:

1. **Range:** The difference between the maximum and minimum values.
2. **Variance:** The average of the squared differences from the mean.
3. **Standard Deviation:** The square root of the variance.
4. **Interquartile Range (IQR):** The difference between the 75th and 25th percentiles.

Inferential Statistics

Work on sample, generalize for population.

A branch of statistics that deals with using data from a sample to draw conclusions about a larger population. It allows you to make predictions and generalisations that go beyond the information contained in the sample itself. For instance; surveys

1. Hypothesis testing:
2. Confidence intervals
3. Regression analysis

5. Statistics Types

Descriptive Statistics

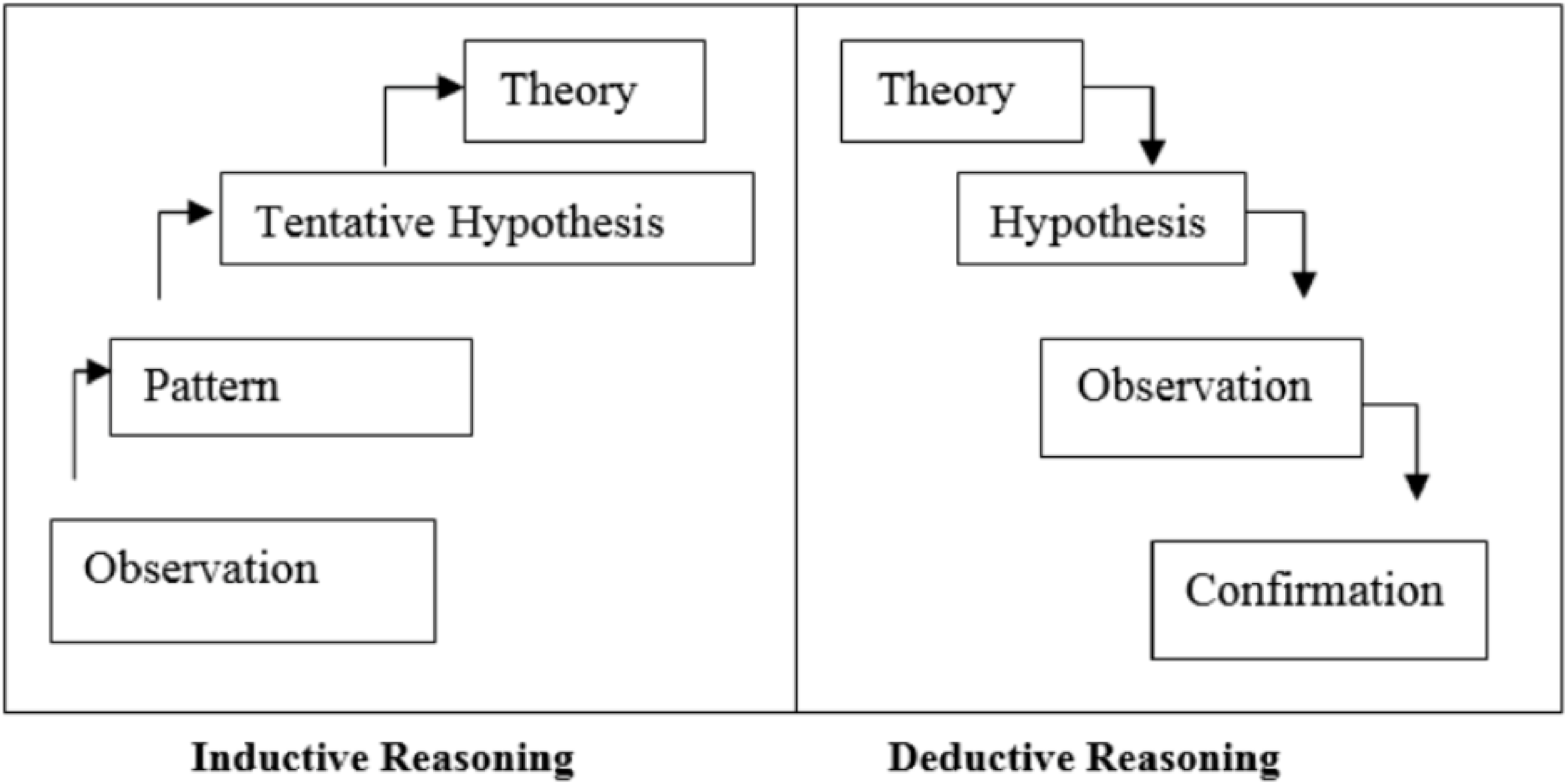
- **Collecting data** on collective events,
- **Process and organize the collected data**,
- **Present organized data in the form of tables or graphs**,
- And ultimately revealing the **trend of collective** events.

Inferential Statistics

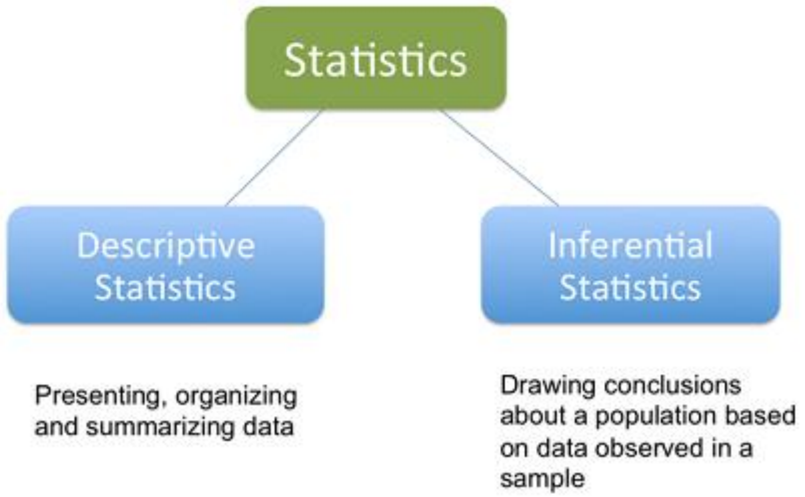
- Based on the concepts of sampling and sampling fractionation, With the help of a **sample selected randomly from a population.**
- **Estimate the parameters of the population**,
- **Testing hypotheses about population parameters**,
- **And making predictions for the future**....
- Since these inferences cannot be absolutely true, they are expressed with the word "**probability**".
- In inferential statistics, the **inductive method** is usually used.

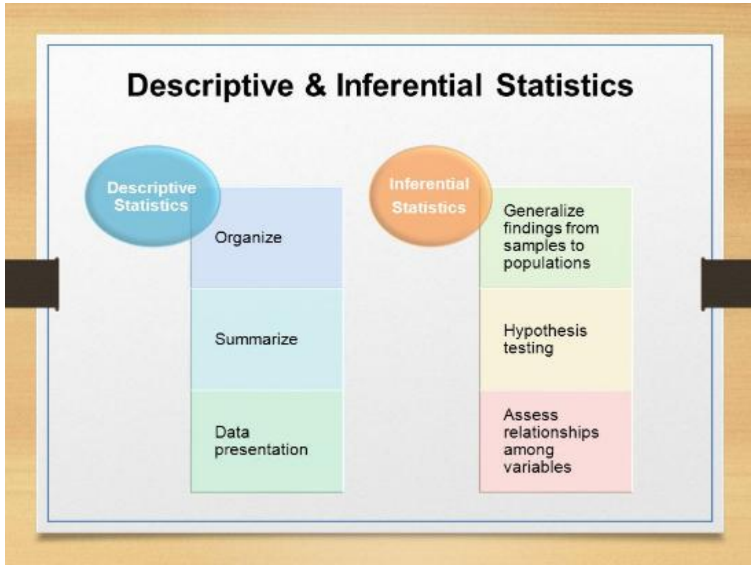
TECHPRO EDUCATION

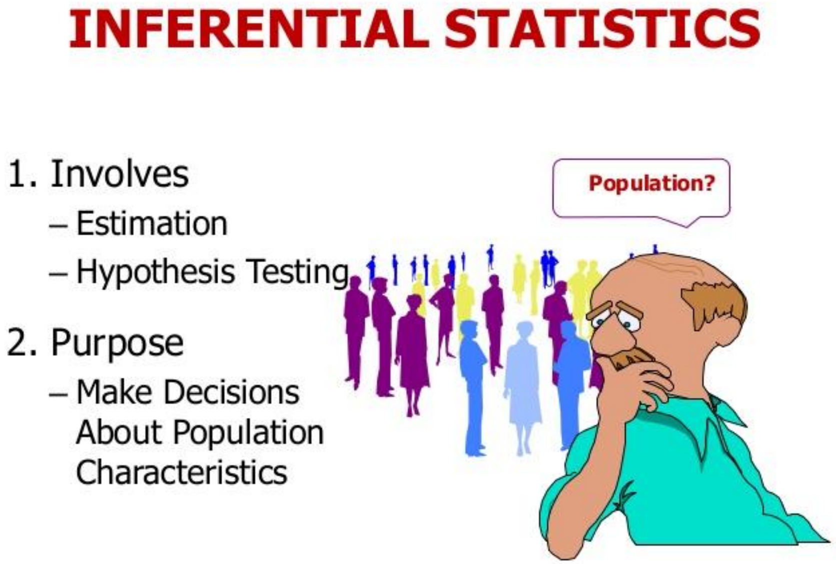
Inductive and Deductive Analyses Algorithms

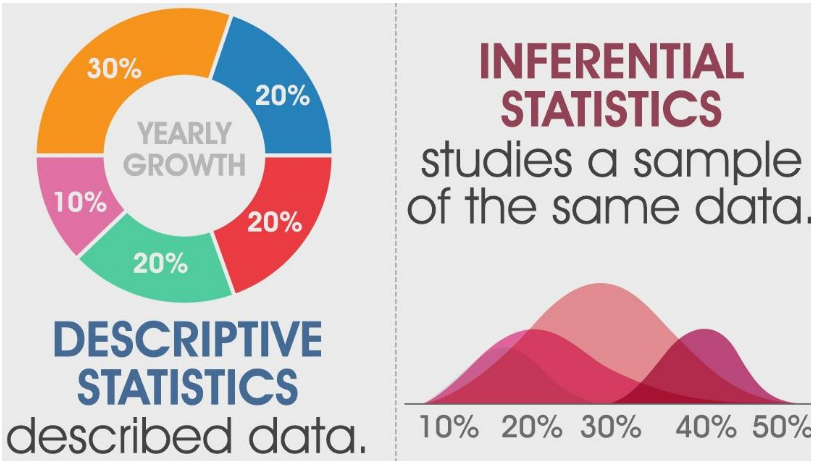


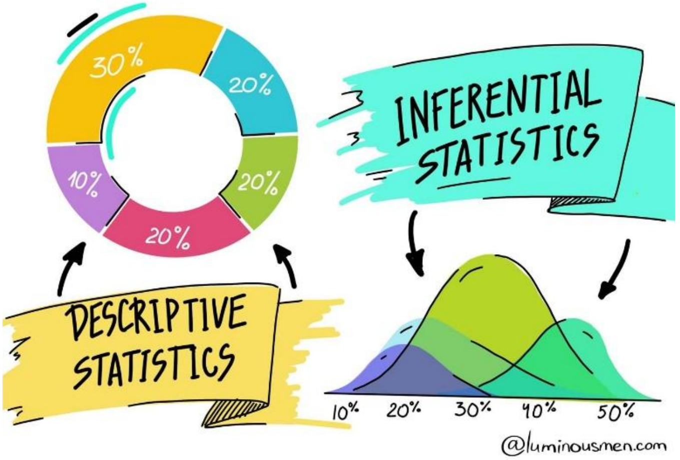
5. Descriptive vs Inferential Statictics







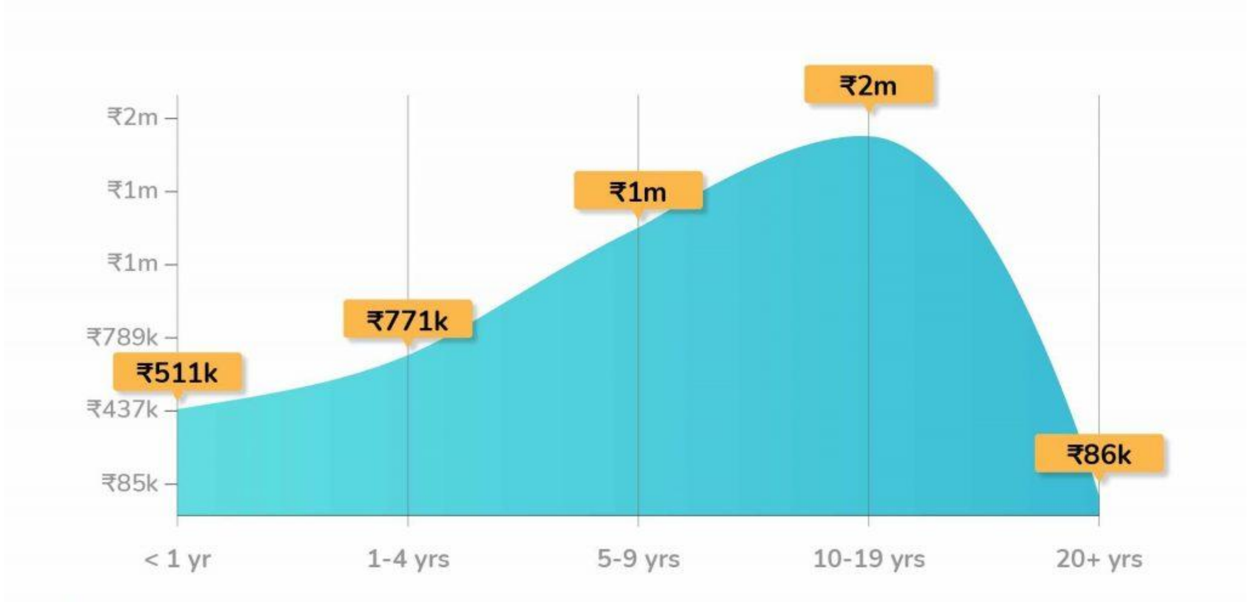


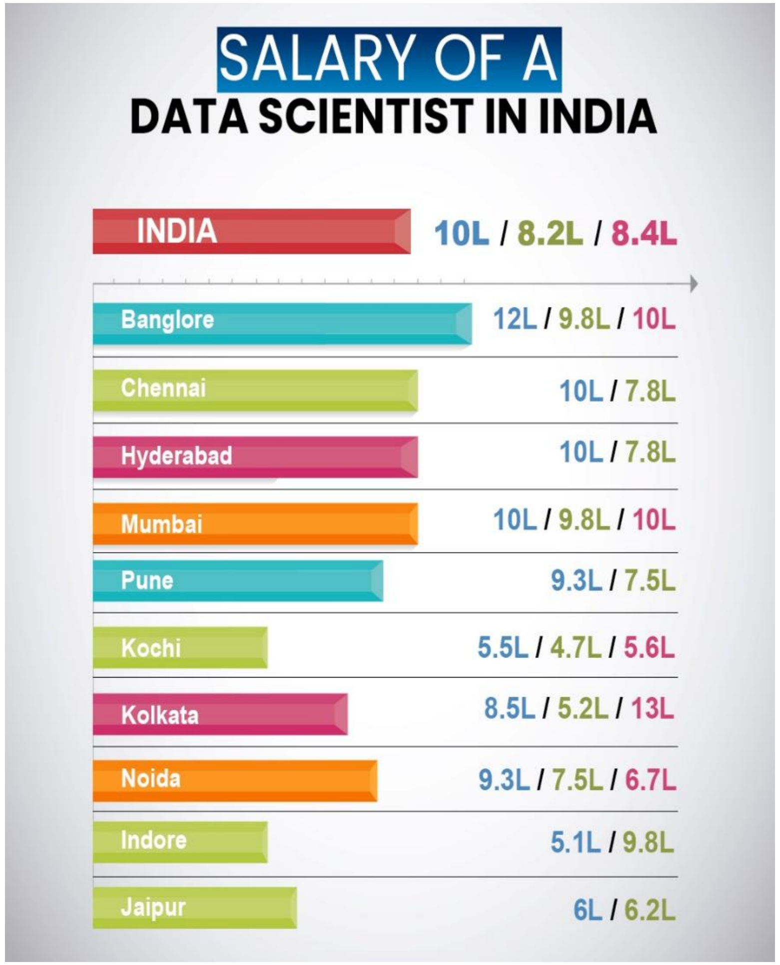


https://www.kaggle.com/code/yashvi/practical-statistics-1-descriptive-statistics

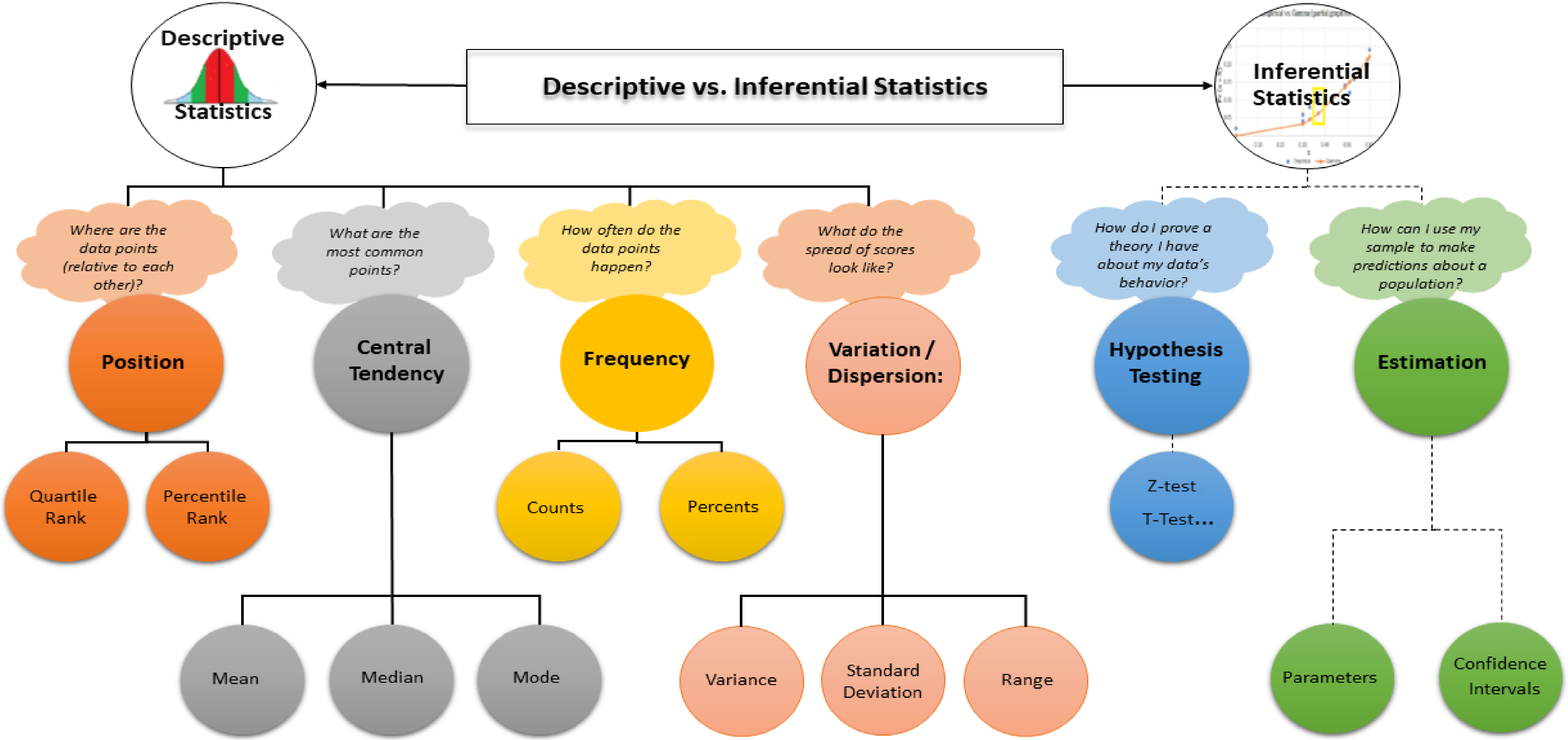
Why do we need Inferential Statistics?

- What is the average salary of a Data Scientist in India?
- **We can use inferential statistics since collecting all data is hard/expensive.**





Descriptive vs Inferential Statictics



Descriptive vs Inferential Statictics

Examples of Descriptive and Inferential Statistics

Descriptive Statistics

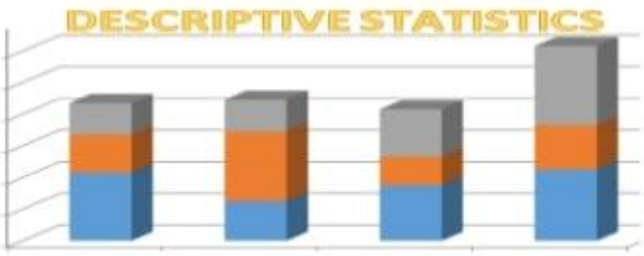
- Graphical
  - Arrange data in tables
  - Bar graphs and pie charts
- Numerical
  - Percentages
  - Averages
  - Range
- Relationships
  - Correlation coefficient
  - Regression analysis

Inferential Statistics

- Confidence interval
- Margin of error
- Compare means of two samples
  - Pre/post scores
  - *t* Test
- Compare means from three samples
  - Pre/post and follow-up
  - ANOVA = analysis of variance

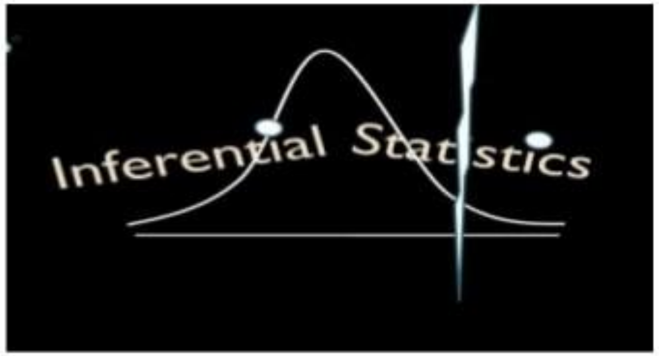
Descriptive Statistics

Descriptive statistics are numbers that are used to summarize and describe data. The word "data" refers to the information that has been collected from an experiment, a survey, a historical record, etc.



Inferential Statistics

The sample is a set of data taken from the population to represent the population. Probability distributions, hypothesis testing, correlation testing and regression analysis all fall under the category of **inferential statistics**.



https://www.youtube.com/watch?v=MUyUaouisZE

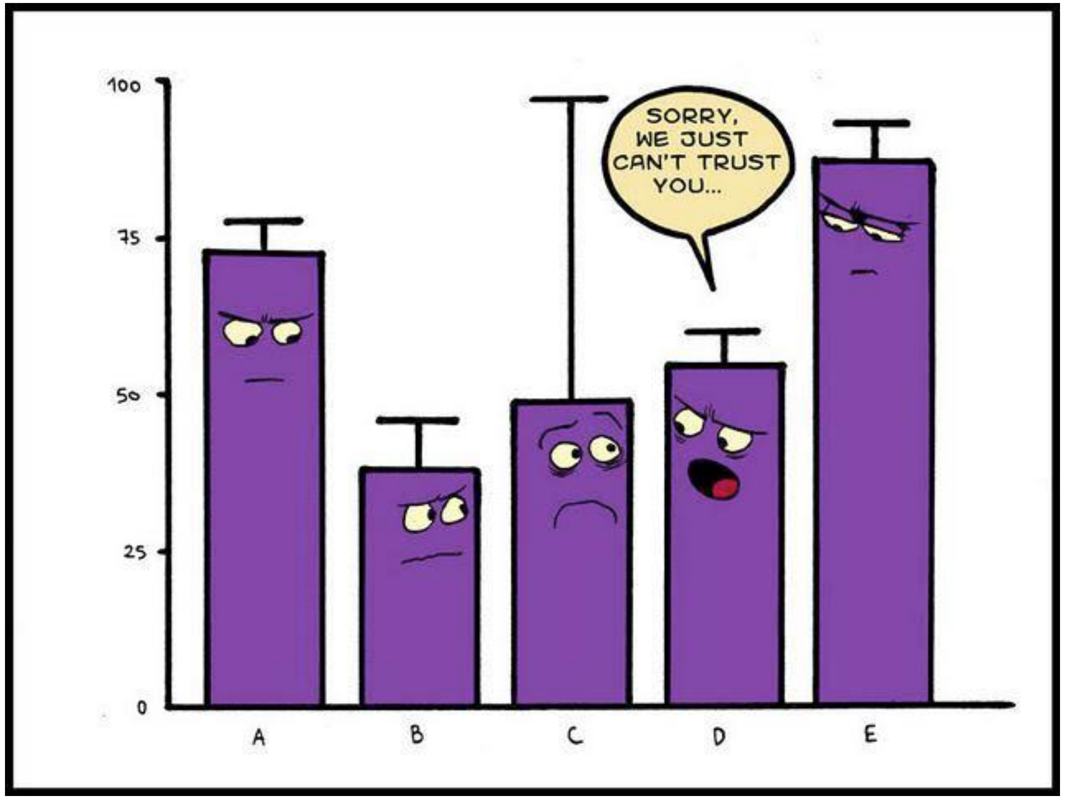
QUESTION

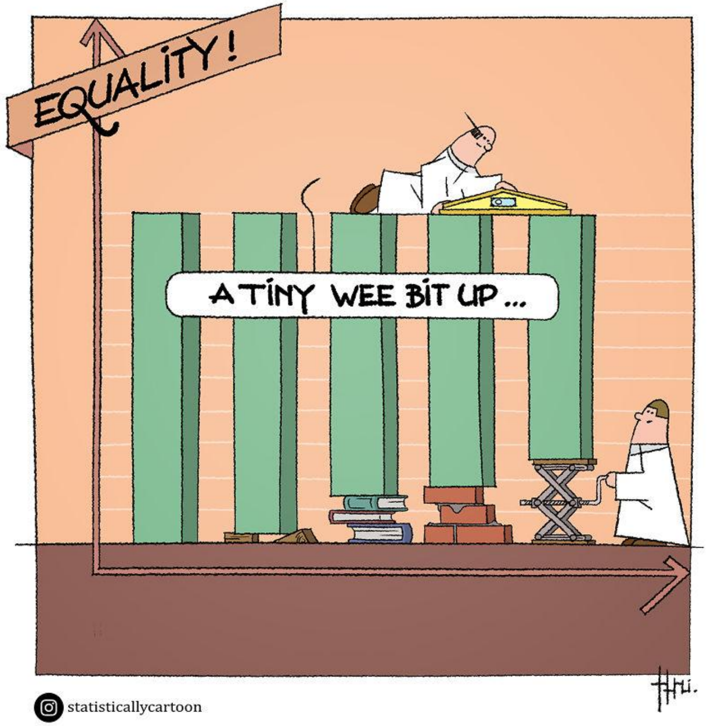
Which is descriptive statistics?

A survey for next election?

Average age of DS-Batch 250 students at TechProEd

Sometimes funny content



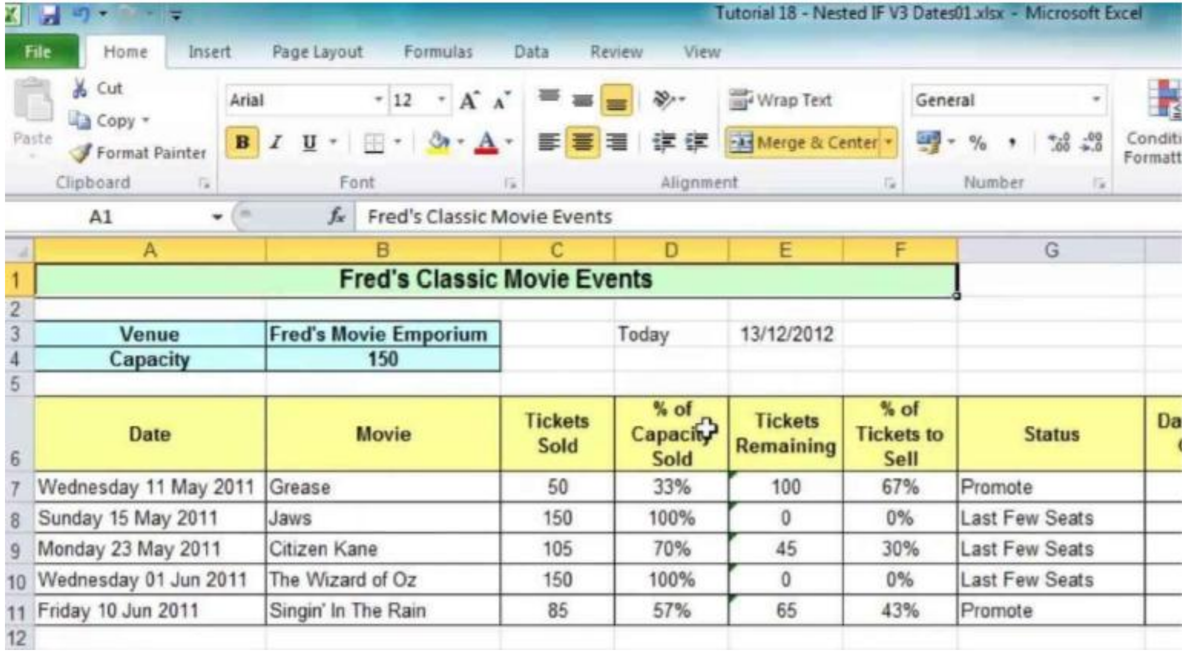


6. Data Types

- ## What is Data?

Data refers to **information**, often facts or **figures**, **amounts**, **labels** (groups, kinds), **colours**, shapes, **images**, **videos**, **voices**, and **text**.

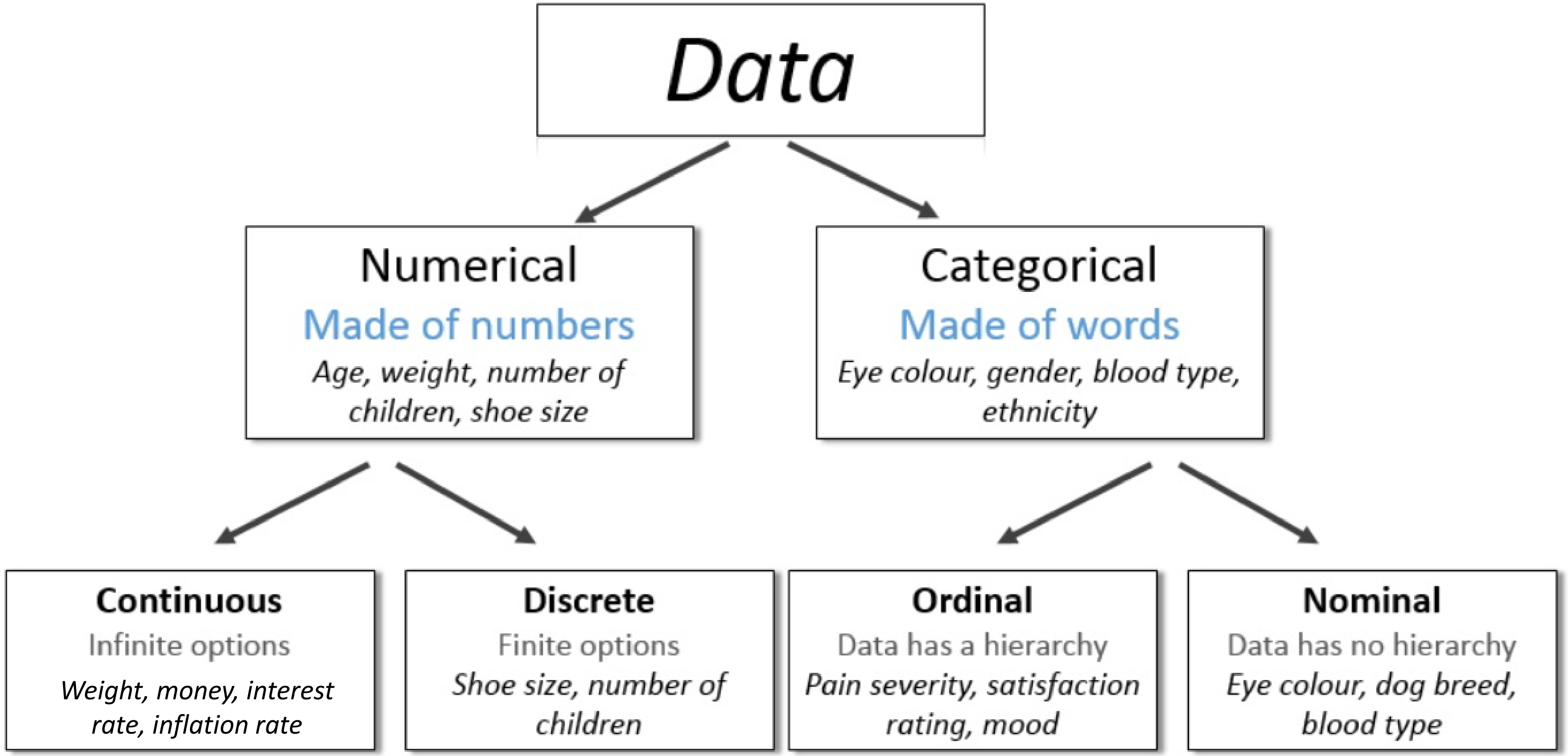
Data is everything.
Posts, messages, emails, shopping information, temperature, vote, prices, inflation, interest rate, GDP, population, ...



Fred's Classic Movie Events

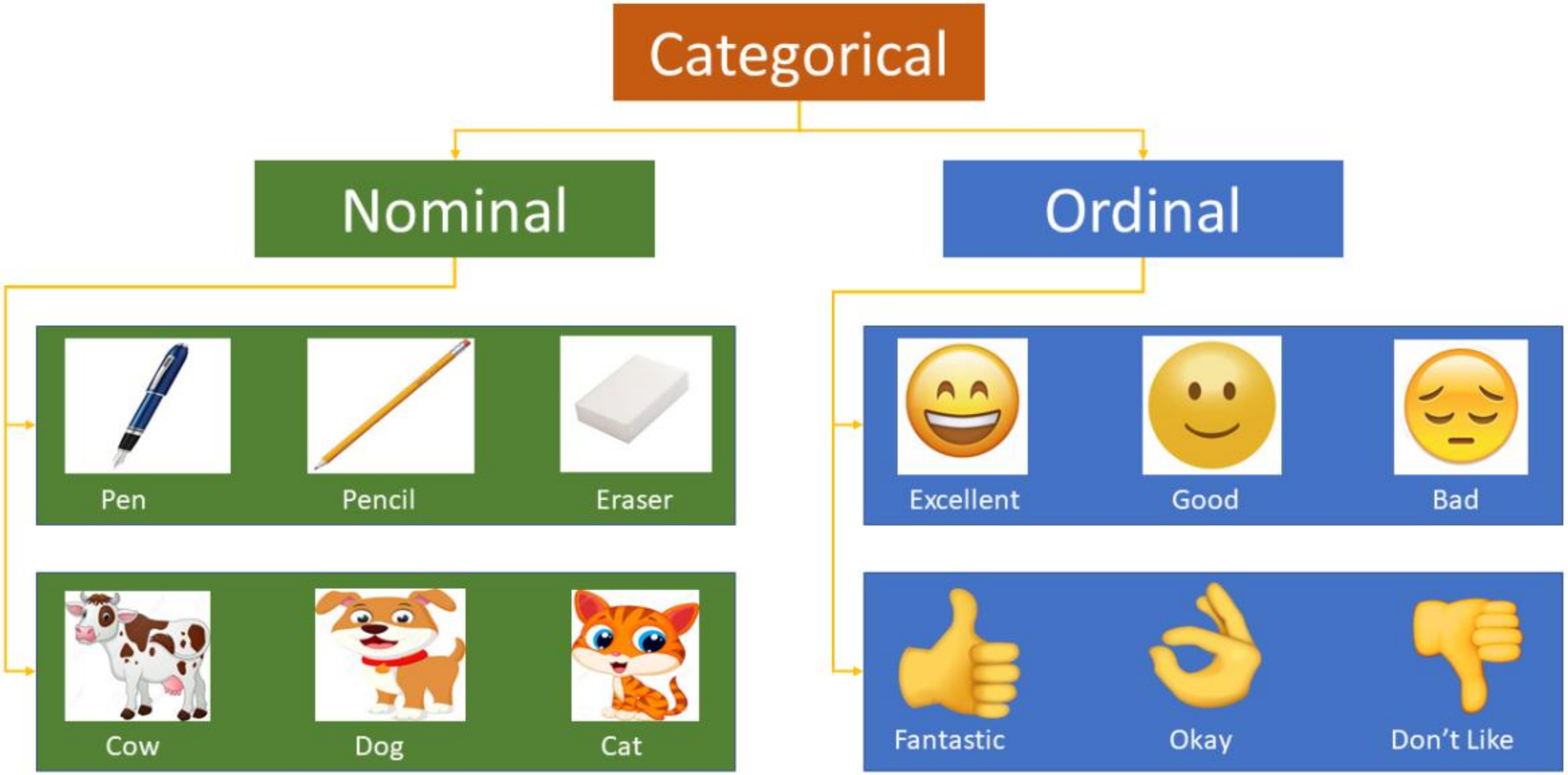
Venue	Fred's Movie Emporium		Today	13/12/2012		
Capacity	150					
Date	Movie	Tickets Sold	% of Capacity Sold	Tickets Remaining	% of Tickets to Sell	Status
Wednesday 11 May 2011	Grease	50	33%	100	67%	Promote
Sunday 15 May 2011	Jaws	150	100%	0	0%	Last Few Seats
Monday 23 May 2011	Citizen Kane	105	70%	45	30%	Last Few Seats
Wednesday 01 Jun 2011	The Wizard of Oz	150	100%	0	0%	Last Few Seats
Friday 10 Jun 2011	Singin' In The Rain	85	57%	65	43%	Promote

6. Data Types



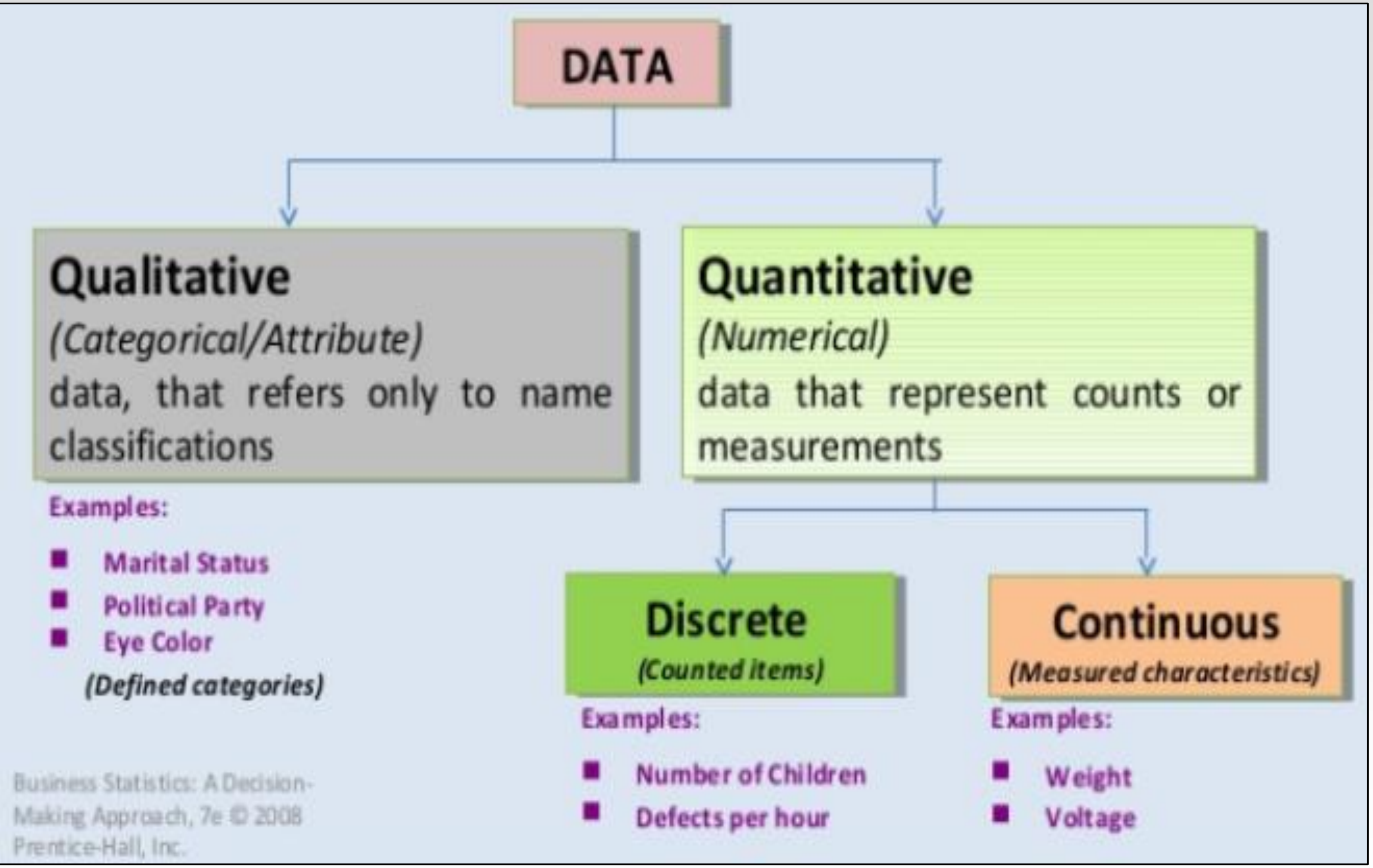
https://www.youtube.com/watch?v=pg12U1BAnoA

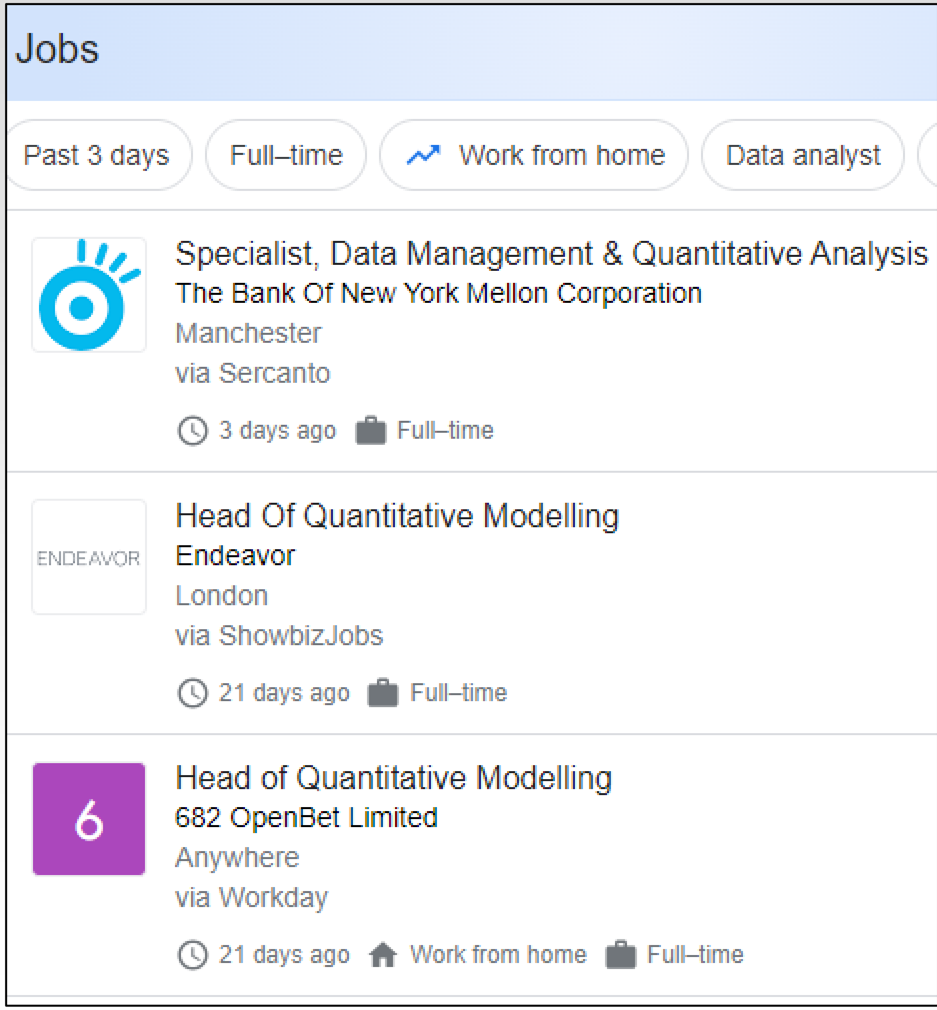
Nominal Data & Ordinal Data



Data Types

If you see a job position named "Quantitative Analysis/Modelling", jump on it. That person is you.





Qualitative

Nominal

- Variables with no inherent order or ranking sequence.
- E.g. Gender, Race etc..

Ordinal

- Variables with an ordered series
- E.g. Blood Group, Performance etc.

Binary

- Variables with only two option.
- E.g. Pass/Fail, Yes/No etc

Quantitative

Discrete

- aka Attribute data
- Discrete data is information that can be categorized into a classification.
- Discrete data is based on counts.
- Finite number of values is possible and the values cannot be subdivided meaningfully.

E.g. – No of Parts damaged in shipment

Continuous

- Continuous data is information that can be measured on a continuum or scale .
- Continuous data can have almost any numeric value and can be meaningfully subdivided into finer and finer increments .
- E.g. – Length, Size, width

Question 1: What are quantitative data and qualitative data?

Answer: Qualitative data is used to describe the characteristics of data and is also known as Categorical data. For example, how many types.

Quantitative data is a measure of numerical values or counts. For example, how much or how often. It is also known as Numeric data.

QUESTION

Which is continous data ?

Race

Height

QUESTION

Which is discrete data ?

Population

Exchange Rate

QUESTION

Which is ordinal data ?

Satisfaction Level (e.g. Low, High)

Bill Amount

QUESTION

Which is nominal data ?

Days of week

Corporate Hierarchy

ML Models Content as Categorical Data

Label Encoder

Raw Data

- We encode the country before exerting it to a ML model

Encoding

Ready for ML/DL Model

- Prepared for ML model

ID	Country	Population
1	Japan	127185332
2	U.S	326766748
3	India	1354051854
4	China	1415045928
5	U.S	326766748
6	India	1354051854

ID	Country	Population
1	0	127185332
2	1	326766748
3	2	1354051854
4	3	1415045928
5	1	326766748
6	2	1354051854

ML Models Content as Categorical Data

One Hot Encoder

Raw Data

- Performing one hot encoding on the country column

ID	Country
1	Japan
2	U.S
3	India
4	China
5	U.S
6	India

Encoding

Ready for Model (ML Model)

- We have **n new** dummy variables (columns) in OneHot Encoder.

ID	Country_Japan	Country_U.S	Country_India	Country_China
1	1	0	0	0
2	0	1	0	0
3	0	0	1	0
4	0	0	0	1
5	0	1	0	0
6	0	0	1	0

ML Models Content as Categorical Data

Dummy Encoder

Raw Data

- Performing dummy encoding on the country column

ID	Country
1	Japan
2	U.S
3	India
4	China
5	U.S
6	India

Encoding

Ready for Model (ML Model)

- We have **n-1** new dummy variables at Dummy Encoder.

ID	Country_Japan	Country_U.S	Country_India
1	1	0	0
2	0	1	0
3	0	0	1
4	0	0	0
5	0	1	0
6	0	0	1

ML Models Content as Categorical Data

Ordinal Encoding

Ordinal Encoding

Grades
A
B
C
D
Fail

Grades	Encoded
A	4
B	3
C	2
D	1
Fail	0

Original Encoding	Ordinal Encoding
Poor	1
Good	2
Very Good	3
Excellent	4

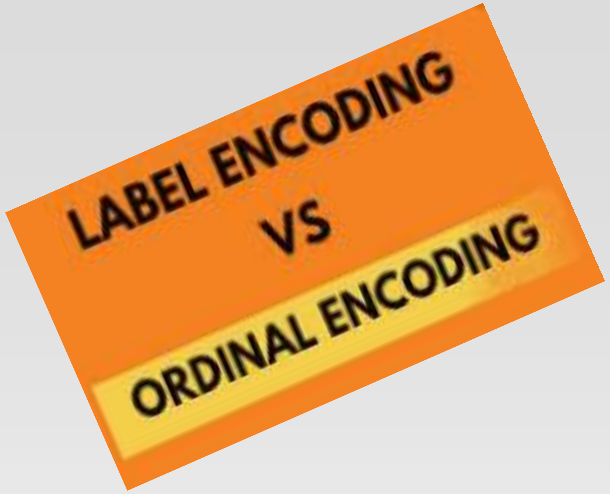
Student	Grade
John	A
Mike	B
Elena	A
Adam	D

Encoding

Student	Grade
John	4
Mike	3
Elena	4
Adam	1

https://www.youtube.com/watch?v=15uClAVV-rI

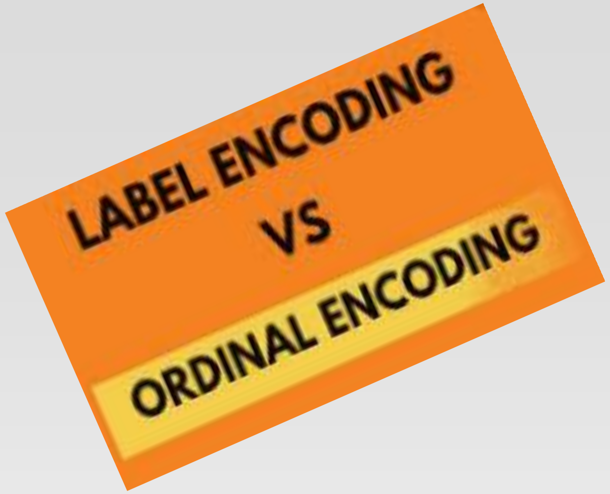
Interview Question from me



Question: What is the difference of Label Encoder and Ordinal Encoder?

Answer: While Label Encoder converts categorical data into numeric and gives numbers from smallest to largest, starting from the alphabetically preceding category, Ordinal Encoder assigns numbers according to the order of importance given by the user.

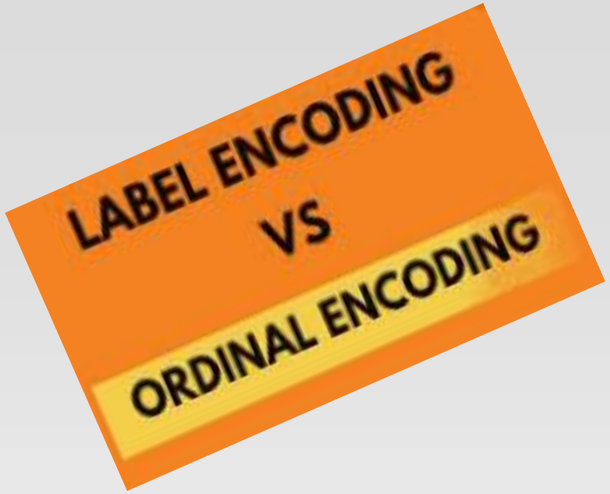
Interview Question from me



Question: What is the difference of One Hot Encoder and Dummy Encoder?

Answer: One Hot Encoding creates binary columns for each category, representing the presence or absence of that category. Dummy Encoding is a specific case of One Hot Encoding where one category is implicitly represented (by dropping its corresponding column) to avoid redundancy in model fitting.

Interview Question from me



Question: Which method(s) is suitable for "Gender"?

A. Label Encoding

B. Ordinal Encoding

C. One Hot Encoding

D. Dummy Encoding

Hands-on Practise Time

StatsModels
Statistics in Python

Seaborn

pandas
$y_{it} = \beta' x_{it} + \mu_i + \epsilon_{it}$

xarray

scikit learn

scikit-image
image processing in python

NumPy

SciPy

matplotlib

jupyter

python™

IP[y]:
IPython

statsmodels

02:00

Start Stop Reset mins: 2 secs: 0 type: None

Breaktime for PowerPoint by Flow Simulation Ltd. Show Settings

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INTERVIEW

HOME WORK

PRACTISE

BEST PRACTISE

NOTE

TASK

TIP

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